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RESEARCH ARTICLE

Interdependence of movement amplitude and tempo during self-paced finger tapping: evaluation of a preferred velocity hypothesis

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Abstract

This study examined the relation between movement amplitude and tempo during self-paced rhythmic finger tapping to test a preferred velocity account of the preferred tempo construct. Preferred tempo refers to the concept that individuals have preferences for the pace of actions or events in their environment (e.g., the desired pace of walking or tempo of music). The preferred velocity hypothesis proposes that assessments of preferred tempo do not represent a pure time preference independent of spatial movement characteristics, but rather reflects a preference for an average movement velocity, predicting that preferred tempo will depend on movement amplitude. We tested this by having participants first perform a novel spontaneous motor amplitude (SMA) task in which they repetitively tapped their finger at their preferred amplitude without instructions about tapping tempo. Next, participants completed the spontaneous motor tempo (SMT) task in which they tapped their finger at their preferred tempo without instructions about tapping amplitude. Finally, participants completed a target amplitude version of the SMT task where they tapped at their preferred tempo at three target amplitudes (low, medium, and high). Participants (1) produced similar amplitudes and tempi regardless of instructions to produce either their preferred amplitude or preferred tempo, maintaining the same average movement velocity across SMA and SMT tasks and (2) altered their preferred tempo for different target amplitudes in the direction predicted by their estimated preferred velocity from the SMA and SMT tasks. Overall, results show the interdependence of movement amplitude and tempo in tapping assessments of preferred tempo.

Keywords Rhythmic tapping · Spontaneous motor tempo · Timing · Motion tracking · Movement trajectories

Introduction

Humans engage in a variety of rhythmic behaviors that range from the simple (periodic) rhythms of walking and hand clapping to the more complex rhythms of dance and music performance (Dahl et al. 2014; Janata et al. 2012; Todd et al. 1999; Tranchant et al. 2016; Van Noorden and Moelants 1999; Wagenaar and Van Emmerik 2000). People vary in the rates (tempi) of these different rhythmic movements, reflecting individual differences in tempo preferences. Measurement of tempo preferences for the generation of different rhythmic movements has led to the concept of *preferred tempo* – defined as the rate (tempo) that is neither too fast nor too slow, but just right for an individual for a given activity.

Across the relatively long history of research on this topic, preferred tempo has been given a variety of different names including ‘mental tempo’, ‘personal tempo’, ‘psychic tempo’ and ‘internal tempo’, each reflecting different assumptions or interpretations about what preferred tempo measures or what it means (e.g., Stern 1900; Wallin 1911; Fraisse 1963, 1982; Jones 1976; McAuley et al. 2006). Perhaps the biggest distinction in interpretation concerns whether preferred tempo only reflects the preferred rate of a

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given motor activity or whether it measures something more general, such as an internal perceptual preference for a particular rate of information flow (Baruch et al. 2004; Boltz 1994; Fraisse 1963, 1967, 1982; Mishima 1956; Moelants 2002; Rimoldi 1951; Stern 1900; Vanneste et al. 2001). A number of studies have supported the latter, more general, interpretation of preferred tempo (see: McAuley et al. 2006; Michaelis et al. 2014; Van Noorden and Moelants 1999).

Preferred tempo has most commonly been measured in the lab using a self-paced rhythmic finger tapping task referred to as spontaneous motor tempo (SMT) task. In the SMT task, participants are asked to tap with their finger or hand at their self-selected, most natural or comfortable rate – a tempo that is not too fast or slow but feels 'just right' for them (Fraisse 1956; McAuley et al. 2006). The recorded data typically consist of the time series of taps with spontaneous (preferred) motor tempo determined by averaging the sequence of produced time intervals between successive taps, resulting in a mean produced inter-tap interval (ITI) measure. In this article, a tap is defined as the moment the tapping finger makes contact with the surface (table) and we will use the terms tempo and inter-tap interval interchangeably. For the SMT task, many studies have reported that preferred tempo, expressed as the mean produced inter-tap interval, is *on average* around 500–600 ms for adults (Fraisse 1982; Hammerschmidt et al. 2021; McAuley et al. 2006; Moelants 2002). When preferred tempo is measured across trials or across days, individuals tend to show relatively high test-retest reliability, but also large individual differences. Sample distributions of preferred tempi obtained using the SMT tasks have a range from around 200 ms at the fast end to 1800 ms at the slow end (Fraisse 1982; Hammerschmidt et al. 2021; Hammerschmidt & Wollner, 2022; Kliger-Amrani and Zion-Golumbic 2020; McAuley et al. 2006).

Studies investigating preferred tempo using the SMT task have identified several factors that may contribute to individual variation in preferred tempo, such as age, arousal, and body size. Preferred tempo has been shown to be faster for children than for adults and further slows in late adulthood (Baudouin et al. 2004; Drake et al. 2000; McAuley et al. 2006; Vanneste et al. 2001). Boltz (1994) reported that increases and decreases in arousal lead to faster and slower preferred tempi, respectively, consistent with the view that arousal modulates the speed of some form of internal clock (see also Dosseville et al. 2002; Hammerschmidt & Wollner, 2022; Moussay et al. 2002). There is also some evidence that larger body size may be associated with slower preferred tempi (Dahl et al. 2014; Todd et al. 2007), but results across studies have been mixed.

One factor that has received relatively little attention in the literature, which has the potential to affect preferred tempo in the SMT task, is movement amplitude. Support for a general relationship between movement amplitude and tempo in the broader movement timing literature comes from a number of sources. Rosenbaum et al. (1991) had individuals make finger, hand, and arm oscillations in the horizontal plane that were either self-paced at different target amplitudes (amplitude driven) or paced by isochronous tone sequences at different tempi leaving movement amplitude unconstrained (tempo driven). They found that individuals' choice of effector (finger, hand, or arm) to complete the task was related to the required amplitude and tempo of the movement. Participants preferentially used smaller effector segments (i.e., finger) for smaller amplitude and faster tempo movements and larger effector segments (i.e., hand or arm) for larger and slower movements. Yu et al. (2003) had participants swing three pendulums of different lengths (36, 40, and 48 cm) at their preferred tempo using wrist movements. Participants preferred slower tempi and larger amplitudes when swinging longer pendulums. Similarly, Kay et al. (1987) had participants synchronize repetitive wrist flexion-extension oscillations with an auditory metronome at a range of tempi. During cyclical wrist movements, changes in movement tempo were associated with corresponding changes in movement amplitude, where amplitude increased at slower movement rates. Ringenbach et al. (2003) had participants draw circles in time with a metronome at tempi between 0.77 and 2.5 Hz, instructing them to complete one circle per metronome cycle. Circle diameter, corresponding to movement amplitude in the drawing task, increased at slower metronome tempi.

Together, these previous studies provide evidence for an interdependence between movement amplitude and movement tempo in the production of simple (periodic) rhythms. Larger and smaller movement amplitudes have been associated with slower and faster produced movement tempi in a variety of contexts, respectively. None of these studies, however, have considered the relation between movement amplitude and movement tempo in the assessment of preferred tempo using the spontaneous motor tempo (SMT) task involving self-paced finger tapping. A main aim of the current study was, thus, to determine the potential interdependence of movement amplitude and movement tempo to the SMT task.

One possibility is that the average produced inter-tap interval associated with the discrete series of finger taps in the SMT task is independent of finger tapping amplitude (an individual's average maximum finger height during performance of the task). This would fit with what is traditionally assumed about the SMT task; it provides a measure of an individual's preferred tempo that is independent of other movement characteristics. Another possibility is that the SMT task does not provide a pure measure of preferred tempo per se, but rather preferred tempo in the SMT task is influenced by tapping amplitude. For this possibility, we specifically considered that the relationship between tapping amplitude and preferred tempo might be governed by individuals having a preferred average velocity rather than a pure tempo preference.

From this perspective, consider that the standard equation for velocity is $V = \Delta S/\Delta T$. For finger tapping, we were concerned with vertical movement, where amplitude (A) represents the peak vertical displacement of the finger from the table between taps, corresponding to half of the total vertical movement distance for one inter-tap interval. In the context of the SMT task, the total movement in space (ΔS) for the total amount of time (ΔT) taken for each inter-tap interval (ITI) corresponds to two times the maximum finger height (amplitude), A. Thus, substituting $2A$ for ΔS, and ITI for ΔT, the equation for velocity can be rewritten as $V = 2A/ITI$ to describe the relation between tapping amplitude and tapping tempo in the SMT task. The term velocity (V) as described here refers to a space/time ratio index of vertical distance and inter-tap interval duration. Thus, from a preferred velocity perspective, individual preferences for finger tapping tempo and finger tapping amplitude should be interdependent. Moreover, increases and decreases in an individual's average finger tapping amplitude would be expected to result in systematic increases and decreases in the average produced inter-tap interval in order to maintain that individual's preferred velocity.

To evaluate these possibilities, participants in the current study completed a set of three self-paced finger tapping tasks that differed in instructions. Participants first performed a novel spontaneous motor amplitude (SMA) task in which they repetitively tapped their finger at their preferred amplitude without any instructions about tapping tempo. Next, participants completed the standard spontaneous motor tempo (SMT) task in which they tapped their finger at their preferred tempo without any instructions about tapping amplitude. Finally, participants completed a target amplitude version of the SMT task where they tapped at their preferred tempo at three target maximum finger heights (low, medium, and high).

If tapping amplitude and tempo are interdependent, then we expected that participants would produce similar amplitudes and tempi in the SMA and SMT tasks independent of whether they were asked to produce their preferred amplitude or their preferred tempo, respectively. Moreover, any systematic difference in produced tempi across tasks should result in parallel changes in produced amplitudes. For example, if preferred tempi speed up or slow down from the first task to the second task then produced amplitude would similarly be expected to decrease or increase in order to maintain a constant preferred velocity. With respect to the target amplitude version of the SMT task, the central question was whether preferred tempi would stay the same across the three target amplitudes or preferred tempi would systematically vary with target amplitude in line with a preferred velocity hypothesis.

A strong version of the preferred velocity hypothesis would predict that produced preferred tempi in the three target amplitude conditions should be a close match to predicted preferred tempi based on the estimated preferred velocity from the SMA and SMT tasks and produced amplitudes in each condition according to the equation: $ITI = 2\,A/V$. However, it is also possible that a weaker version of the preferred velocity hypothesis holds whereby estimated velocity from the SMA and SMT tasks constrains preferred tempi, but the predicted and produced preferred tempi do not precisely match. That is, lower and higher target amplitudes lead to faster and slower preferred tempi, respectively, but the amount of influence velocity has on preferred tempo is not absolute and could differ from individual to individual.

Methods

Participants and design

Twenty undergraduates (15 female, 5 male) between the ages of 18 and 26 years ($M = 19.8$, $SD = 2.3$) from the Michigan State University community completed the experiment for partial credit in an undergraduate psychology course. Based on self-report, 17 participants were right-handed, two were left-handed, and one was ambidextrous. Participants completed all tasks with their self-reported dominant hand (the ambidextrous participant completed the tasks with their right hand). All participants completed a maximum tapping amplitude task followed by three unpaced rhythmic finger tapping tasks: a spontaneous motor amplitude (SMA) task, a spontaneous motor tempo (SMT) task, and a target amplitude version of the SMT task. The order of tasks was fixed so that measures of SMA and SMT were not influenced by previously tapping at different target amplitudes. Prior to participation, all participants gave informed consent; the study was approved by the Michigan State University Institutional Review Board.

Stimuli and apparatus

Tapping movements were recorded using an electromagnetic motion tracking system (trakSTAR by Ascension Technology Corp, Shelburne, VT, USA) at a 200-Hz sampling rate. One sensor from this continuous motion-tracking device was attached to the distal phalanx of the index finger of the participant's tapping hand, providing x, y, and z coordinates of finger movements. For the target amplitude version of the SMT task, target amplitudes were presented using an amplitude guide on the table in front of the participant. This guide was made of wooden planks with 2 digital calipers on each side set to move each adjustable arm in the vertical plane (Fig. 1). A thin black elastic thread connected the moving arms of the calipers, spanning them in the horizontal plane, which allowed target amplitudes to be set by moving the arms of both calipers up or down. The calipers displayed a readout of the height of the string in mm, where the baseline of 0 mm was set at the point where the elastic thread was barely touching the top of the sensor on the participants' tapping finger when at rest on the table. This made it possible to account for differences in the thickness of each participant's index finger. To set each target amplitude, the experimenter manually adjusted each caliper arm from the 0 mm baseline to the target height. The elastic thread functioned as the target amplitude stimulus, which was positioned directly above the individual's tapping finger. Target amplitude was reached when the elastic was visibly disturbed without being displaced far enough to create a perceptible resistance (i.e., not stretched to the point that it slowed finger movement), ensuring that participants could not rely on proprioceptive feedback to match the target amplitude.

Procedure

Participants performed four tasks in the following fixed order: (1) a maximum amplitude task in which participants raised their finger as high as possible off the Table (2 trials), (2) a spontaneous motor amplitude (SMA) task in which participants tapped their finger at a comfortable finger height (amplitude) that was neither too high nor too low

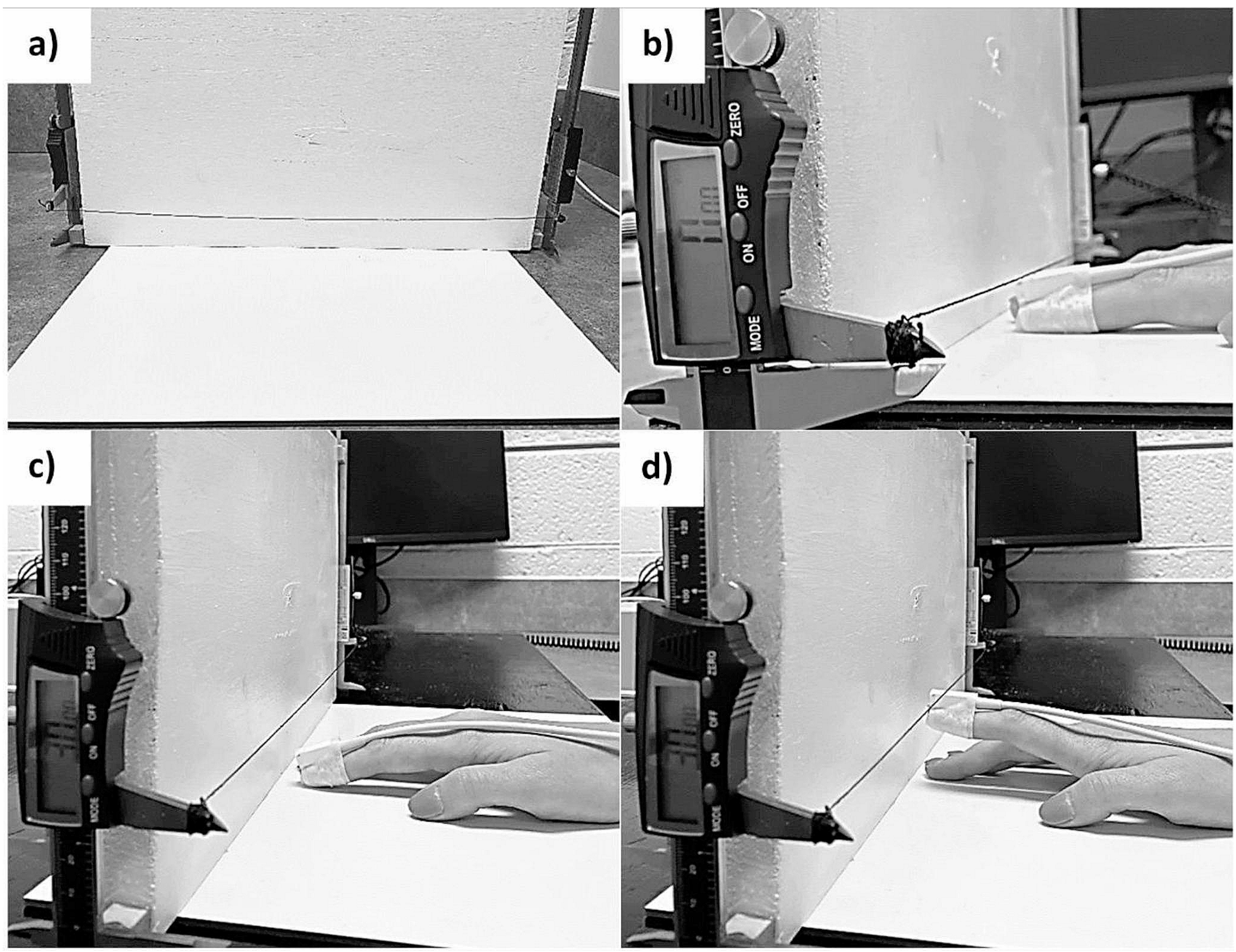


Fig. 1 Photographs of the amplitude guide apparatus. (**a**) Amplitude guide with elastic thread spanning adjustable caliper arms on each side. (**b**) Amplitude 0 calibration: calipers were each set to 0 mm at the location corresponding to the top of the sensor attached to the participant's tapping finger with medical tape. (**c**) Example of 30 mm target amplitude with finger at tap. (**d**) Example of 30 mm target amplitude with finger at target amplitude

but just right for them, with no instructions about tapping tempo (4 trials, 40 s of tapping per trial), (3) a spontaneous motor tempo (SMT) task in which participants tapped at a comfortable tempo that was neither too fast nor too slow for them, with no instructions about tapping amplitude (4 trials, 40 s of tapping per trial), and (4) a target amplitude version of the SMT task in which participants tapped at their preferred comfortable tempo at three target amplitudes (10 mm, 30 mm, 50 mm) where there were 4 trials at each target amplitude and 40 s of tapping per trial.

For the maximum amplitude task, participants wore headphones and started with their hand relaxed and resting on the table. They followed recorded instructions to lift their index finger as high as possible off the table and hold it there (~ 5 s), then to relax their finger back to the table; they repeated that process once during the trial and completed two trials of the maximum amplitude task (each trial was ~ 40 s with instructions). The mean of the four amplitudes produced across the two trials was used as the estimate of each participant's maximum tap amplitude.

For each of the four 40-sec trials of the SMA task, participants tapped a regular beat at their comfortable (preferred) finger height. They were instructed to choose a finger height between taps that was "not too high and not too low, but just right for them" while maintaining a regular beat (with no instructions about tapping tempo). The mean produced tap amplitude across the four trials was used to estimate each participant's preferred amplitude.

For each of the four 40-sec trials of the SMT task, participants tapped a regular beat at their comfortable tapping tempo. They were instructed to choose a tempo that was "not too fast and not too slow, but just right for them" and maintain a steady beat throughout (with no instructions for tapping amplitude). The mean produced inter-tap interval across the 4 trials was used to estimate each participant's preferred tempo.

For each of the twelve 40-sec trials of the target amplitude version of the SMT task (4 trials at each of the three target amplitudes), participants were instructed to tap at their preferred comfortable pace, "not too fast and not too slow, but just right for them" while also matching the height of their finger between taps to the height of the string on the amplitude guide. Participants placed their hand on the amplitude guide apparatus with their tapping finger positioned directly below the target amplitude string. They raised their finger high enough for the motion sensor (fixed on top of their tapping finger) to barely touch the string above before lowering their finger back to the table. Their goal was to just noticeably vibrate the string between each tap while maintaining a steady tapping beat at their most comfortable tempo.

Before setting each target amplitude, the experimenter moved both caliper arms into position so that the string just barely touched the sensor on top of the participant's finger while at rest on the table. This was set as the 0 position so that each target height was exactly 10, 30, or 50 mm above the neutral resting point of the participant's finger on the table. Because the string on the amplitude guide was made of very thin elastic and the motion-tracking sensor was on top of the tapping finger, participants could not feel when they met the target amplitude and had to look at their tapping finger, meaning that amplitude matching was visually-guided and there should have been minimal reliance on tactile feedback. The mean produced inter-tap interval across the four trials at each target amplitude was used to estimate each participant's preferred tempo at each of the three target amplitudes.

After completing the four tasks, participants completed a demographic survey that included questions about age, sex, ethnicity, handedness, music training and experience, language background, hearing ability, general health and wellbeing on the day of the experiment, and whether any strategies were used for completing the experiment. Finally, we measured finger length and hand size, and recorded participants' self-reported height. The entire experiment lasted ~ 60 min.

Data processing and analysis

For each trial, we extracted the time series of taps and amplitudes from the continuous finger data using a customized MATLAB script with the Signal Processing Toolbox® (The MathWorks, Inc., MA, USA). Figure 2 shows an exemplar of raw continuous finger motion data, denoting the three main variables of interest: time of tap, the associated inter-tap interval and tap amplitude.

The primary dependent measures were the produced tap amplitudes and produced inter-tap intervals (ITIs) for the spontaneous motor amplitude (SMA) task, spontaneous motor tempo (SMT) task, and SMT task with three target amplitudes averaged over the 4 trials of each task. Produced amplitudes and produced ITIs were also combined for each participant to create a tap velocity measure ($V = 2\ A/ITI$) for the SMA and SMT tasks that provided an estimate of average vertical movement distance over inter-tap interval for each task. Because trial duration was fixed at 40 s and tapping tempo varied across participants, the number of produced inter-tap intervals also varied across participants. To equate the number of produced inter-tap intervals across participants, data analysis examined only the first 20 produced inter-tap intervals on each trial. Trials with fewer than 20 inter-tap intervals were removed, which resulted in the elimination of 5 out of a total of 400 tapping trials (1.25%) across the 20 participants.

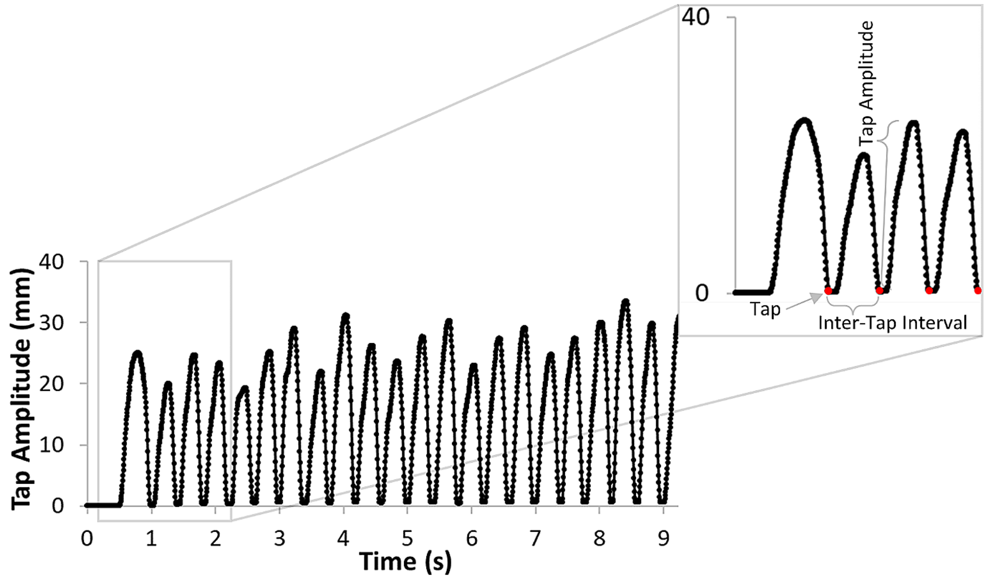


Fig. 2 Example of continuous finger motion data showing one SMT trial from one participant. The line shows the trajectory of the finger over time, where each point represents the vertical position sampled every 5 ms (200 Hz sampling rate). Our MATLAB program extracted the taps (the z-axis minimum positions corresponding to the location of the sensor when the index finger made contact with the table) and the time series of tap amplitudes (the peak height of the index finger between taps). Local minima (at 0 amplitude) are finger taps (red dots) and local maxima are tap amplitudes. Key dependent variables are labeled in the enlarged plot showing the first few taps

Results

Maximum tap amplitude

First, we examined maximum produced amplitudes in the maximum tap amplitude task to establish the maximum finger heights that participants could produce. Overall, maximum possible amplitudes produced by participants varied between 43.8 and 86.8 mm ($M = 64.5$ mm, $SD = 12.0$ mm). Of note, the maximum possible tap amplitudes for all but one participant exceeded the 50 mm target amplitude in the target amplitude version of the SMT task supporting the use of 50 mm for the highest amplitude in the target amplitude version of the SMT task. Because individual differences in range of motion may be influenced by biomechanical factors such as finger length and connective tissue elasticity (e.g., superficialis tendons, extensor tendons), we also considered the possibility that finger length might predict maximum tap amplitude. At least for the range of finger lengths of the participants in the present study, a Pearson correlation analysis showed no reliable relationship between finger length and maximum tap amplitude, $r(18) = 0.28$, $p = 0.23$.

SMA and SMT tasks

Both the SMA and SMT tasks yielded measures of tapping amplitude and tempo, but for the SMA task, produced amplitudes assessed participants' preferred amplitude of finger movement with no instructions given about tapping tempo, while for the SMT task, produced tempi assessed participants' preferred tempo of finger movement with no instructions given about tapping amplitude. A descriptive comparison of produced tempi and amplitudes across the two tasks revealed that participants produced similar but slightly faster preferred tempi (shorter produced inter-tap intervals) in the SMT task ($M = 556$ ms, $SD = 230$ ms) than in the SMA task ($M = 610$ ms, $SD = 200$ ms) and produced similar but slightly lower amplitudes in the SMT task ($M = 29.2$ mm, $SD = 11.0$, range = 13.0–55.3) than in the SMA task ($M = 32.0$ mm, $SD = 9.8$ mm range = 12.7–48.1). The observed differences across tasks, however, only approach significance for both produced tempo ($t(19) = 1.77$, $p = 0.09$, 95% CI = -10.0–117.6; $d = 0.39$) and produced amplitude, $t(19) = 1.83$, $p = 0.08$, 95% CI = -0.4–5.9; $d = 0.41$. To further characterize performance on the novel SMA task, we investigated potential relations between mean amplitude on this task and participants' maximum possible amplitude, finger length, as well as potential sex differences. Pearson correlations showed a relationship between preferred amplitude and maximum possible amplitude that bordered on statistical significance, $r(18) = 0.436$, $p = 0.055$, but no relationship between preferred amplitude and finger length $r(18) = 0.11$, $p = 0.65$. A two-tailed independent samples t-test showed no difference between preferred amplitudes produced by males and females: $t(18) = 0.798$, $p = 0.435$, $d = 0.412$.

Regarding the relationship between produced amplitudes and tempi across the two tasks, one possibility is that despite the similar means produced in each task, preferred amplitudes and preferred tempi in the SMA and SMT tasks, respectively, will be unrelated to (not correlated with) the corresponding produced amplitudes and tempi in the SMT and SMA tasks. If, however, produced amplitudes and tempi are constrained by a preferred velocity (estimated by *2 A/ITI*), then one would expect produced amplitudes and produced tempi to be positively correlated, respectively, across tasks, independent of whether participants were asked to produce their preferred amplitude or their preferred tempo.

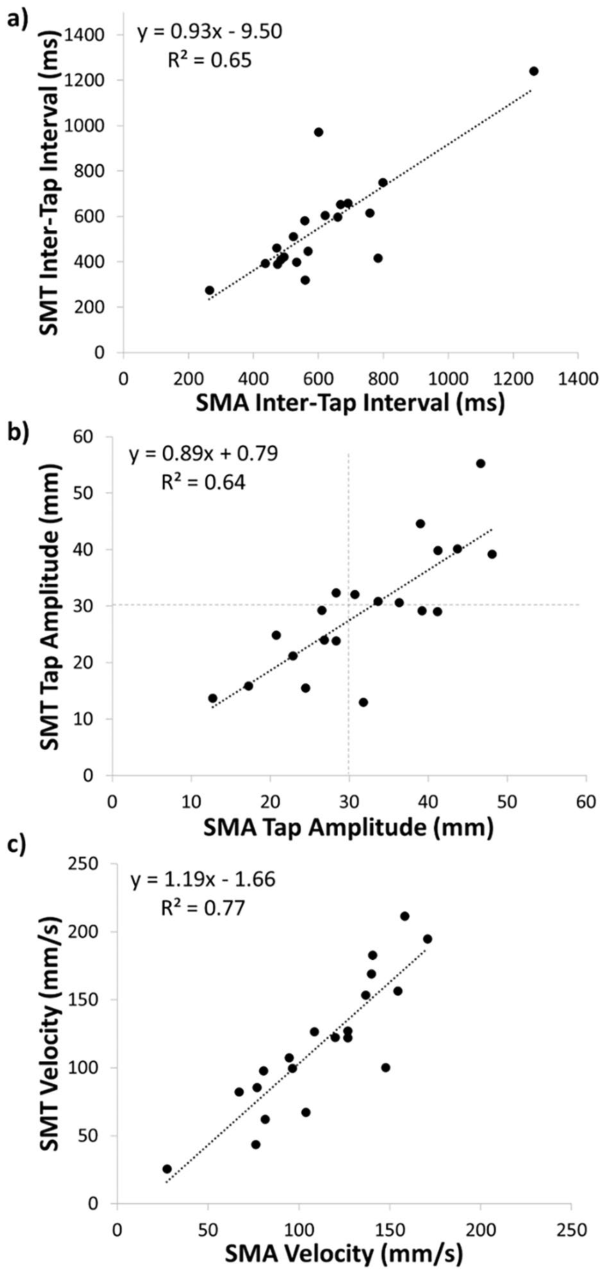


Fig. 3 Scatter plots with best linear fit showing relation between produced inter-tap intervals (ITIs) in the SMA and SMT tasks (Panel **A**), relation between produced amplitudes in the SMA and SMT tasks with reference lines at 30 mm (Panel **B**), and relation between estimated velocities in the SMA and SMT tasks (Panel **C**)[1]

[1] To address whether the outlier in Fig. 3a may have affected the results, we calculated Spearman's rank-ordered correlation coefficient rho for the data reported in both Fig. 3a and b. For the correlation in Panel A: $\rho(18)=0.74$, $p<0.001$, and for the correlation in Panel B: $\rho(18)=0.74$, $p<0.001$

Moreover, velocity (combining the mean amplitude and mean tempo produced by each participant in each task using *2 A/ITI*) should be approximately the same for each participant in the SMA and SMT tasks.

Consistent with the latter preferred velocity hypothesis, produced tempi were positively correlated across the SMA and SMT tasks ($r(18)=0.81$, $p<0.001$) and produced amplitudes were positively correlated across the SMA and SMT tasks ($r(18)=0.80$, $p<0.001$). Figure 3 (Panels A and B) contains scatterplots showing these relationships and the corresponding best linear fits. Moreover, a comparison of estimated velocities in the two tasks shows that mean estimated velocities were not significantly different and nearly identical (SMA task: $M=11.2$ cm/s, $SD=3.7$; SMT task: $M=11.7$ cm/s, $SD=5.0$), $t(19)$ = -0.940, $p=0.36$, 95% CI = -1.70–0.65, d = -0.21. Moreover, a linear regression using estimated velocity in the SMA task to predict estimated velocity in the SMT task accounts for 77% of the variance with an intercept that was not significantly different from zero (intercept = -0.017, 95% CI = -0.055–0.021) and a slope that was close to one (slope = 1.19, 95% CI = 0.87–1.52) showing that on a participant-by-participant basis estimated velocities across the two tasks were very similar and even in a number of cases identical; see Fig. 3 (Panel C).

SMT task – target amplitude version

Next, we examined the effect of different target amplitudes on preferred tempo. If preferred tempo reflected an individual's preferred velocity, then setting different target amplitudes should systematically alter participants' produced preferred tempo. Specifically, lower and higher target amplitudes should result in faster (shorter) and slower (longer) preferred tempi (produced ITIs), respectively.

With respect to target amplitude matching, Fig. 4A shows that participants, as instructed, produced the lowest amplitudes in the 10 mm target amplitude condition ($M=18.1$ mm, $SD=3.1$ mm), intermediate amplitudes in the 30 mm target amplitude condition ($M=37.0$ mm, $SD=3.0$ mm) and the highest amplitudes in the 50 mm target amplitude condition ($M=56.3$ mm, $SD=4.3$ mm). A trend analysis revealed a significant linear increase in produced amplitudes as a function of target amplitude, $F(1,19)=2839.2$, $p<0.001$, with significant differences in produced amplitudes between the 10 and 30 mm target amplitude conditions, $t(19)=38.8$, $p<0.001$, 95% $CI=17.9$–19.9, $d=8.7$, between the 30 and 50 mm target amplitude conditions, $t(19)=28.8$, $p<0.001$, 95% $CI=17.9$–20.7, $d=6.4$, and between the 10 and 50 mm target amplitude conditions, $t(19)=53.3$, $p<0.001$, 95% $CI=36.7$–39.7, $d=11.9$.

With respect to preferred tempo, Fig. 4B shows that produced ITIs were shortest (preferred tempi fastest) for the

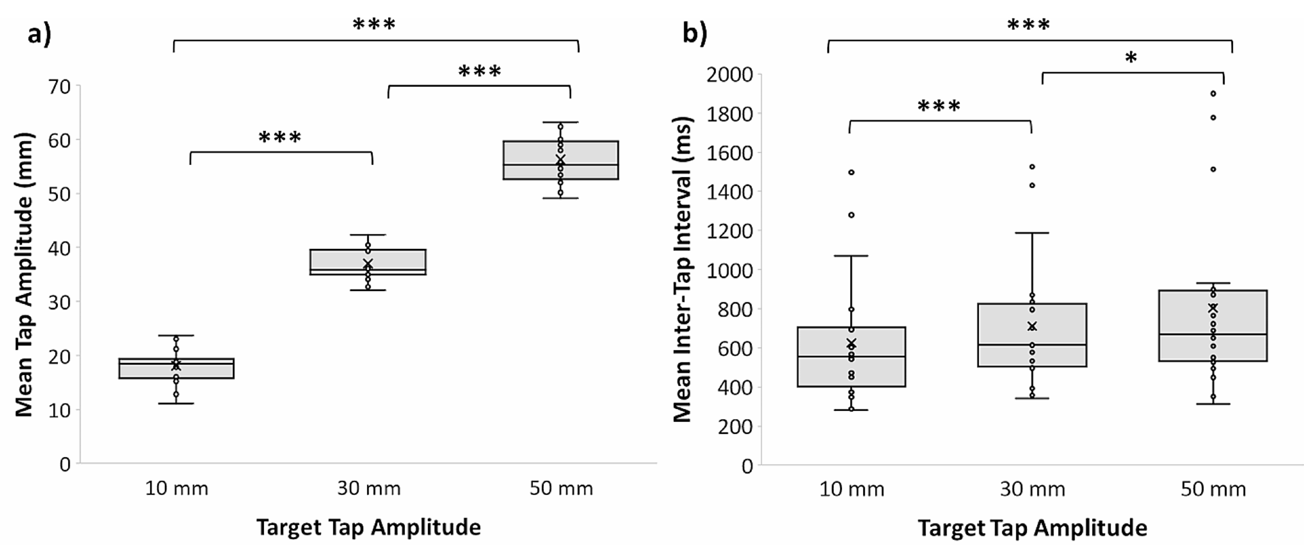


Fig. 4 Box and whisker plots showing produced tap amplitudes (Panel **A**) and preferred inter-tap intervals (Panel **B**) for each participant for the 10 mm, 30 mm, and 50 mm target amplitude versions of the SMT task. X's are group means. * $p<0.05$,** $p<0.01$, *** $p<0.001$

shortest, 10 mm, target amplitude condition ($M=625$ ms, $SD=321$ ms), intermediate for the 30 mm target amplitude condition ($M=710$ ms, $SD=327$ ms) and longest (preferred tempi slowest) for the 50 mm target amplitude condition ($M=803$ ms, $SD=437$ ms). As with the amplitude analysis, a trend analysis revealed a significant linear increase in produced inter-tap intervals as a function of target amplitude, $F(1,19)=15.2$, $p<0.001$, with significant differences in preferred tempi between the 10 and 30 mm target amplitude conditions, $t(19)=4.02$, $p<0.001$, 95% $CI=39.3$–124.5, $d=0.90$, between the 30 and 50 mm target amplitude conditions, $t(19)=2.12$, $p=0.047$, 95% $CI=1.2$–184.7, $d=0.47$, and between the 10 and 50 mm target amplitude conditions, $t(19)=3.9$, $p<0.001$, 95% $CI=81.1$–268.8, $d=0.87$. There was some individual variation in the direction of change in preferred tempo across target amplitudes: 17 out of 20 participants (85%) showed a slowing of preferred tempo from 10 to 50 mm target amplitudes (the magnitude of change varied over a wide range from ~8 to 600 ms). For the remaining three participants (15%) who showed a speeding of preferred tempo from 10 to 50 mm, the amount of tempo change was very small (<1 ms to ~40 ms), indicating that preferred tempo remained the same across the three target amplitudes for those three participants.

Finally, we created a composite estimate of preferred velocity, V, by averaging the velocity estimates from the SMA and SMT tasks (which we earlier showed were very similar on a participant-by-participant basis) and combined V with the produced amplitude, A, in the 10 mm, 30 mm, and 50 mm amplitude conditions (using the formula $ITI=2\,A/V$) to generate a predicted inter-tap interval for each participant in each target amplitude condition. We then (1) examined the correlations between predicted and produced tempi in each condition, (2) examined the correlations between preferred velocities and observed velocities in each condition, and (3) compared the mean estimated preferred velocities from the SMT and SMA tasks with the mean observed velocities in the three target amplitude conditions.

There was a robust positive correlation between predicted tempo and produced tempo in the 10 mm target amplitude condition, $r(18)=0.71$, $p<0.001$, 30 mm target amplitude condition, $r(18)=0.67$, $p=0.001$, and 50 mm target amplitude condition, $r(18)=0.57$, $p=0.009$, indicating that participants changed their tapping tempo in the direction predicted by their estimated preferred velocity and produced amplitude (Fig. 5). There were also significant positive correlations between preferred velocities and observed velocities in all three target amplitude conditions; 10 mm: $r(18)=0.61$, $p=0.004$, 30 mm: $r(18)=0.56$, $p=0.01$, and 50 mm: $r(18)=0.64$, $p=0.003$ (Fig. 6).

Paired-samples t-tests showed that although mean observed velocity in the 30 mm target amplitude condition ($M=12.3$ cm/s, SD $=4.8$ cm/s) was not significantly different from mean estimated preferred velocity ($M=11.4$ cm/s, SD $=4.3$ cm/s), $t(19) = -0.89$, $p=0.39$, 95% $CI = -2.8$–1.1, $d = -0.19$, mean observed velocity was significantly faster than preferred velocity in the 10 mm target amplitude condition ($M=7.1$ cm/s, $SD=3.5$ cm/s) $t(19)=5.53$, $p<0.001$, 95% $CI=2.6$–5.9, $d=1.24$ and significantly slower than preferred velocity in the 50 mm target amplitude ($M=17.4$ cm/s, $SD=8.1$ cm/s), $t(19) = -4.23$, $p<0.001$, 95% $CI = -8.9$–3.0, $d = -0.95$. Thus, although mean produced inter-tap intervals (preferred tempi) shifted in the predicted direction with changes in target amplitude, predicted velocity was an over-estimate of velocity (too fast) in the 10 mm target amplitude condition and an underestimate of

Fig. 5 Scatter plots with best linear fit showing relation between produced SMTs (mean ITIs) and predicted SMTs based on estimated preferred velocity for the three target amplitude conditions

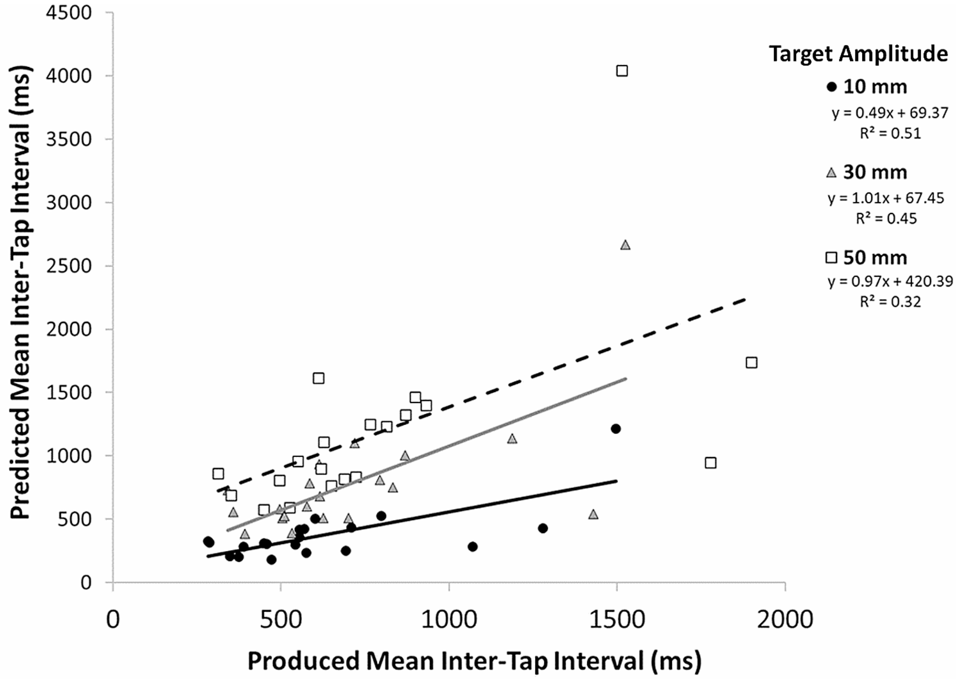
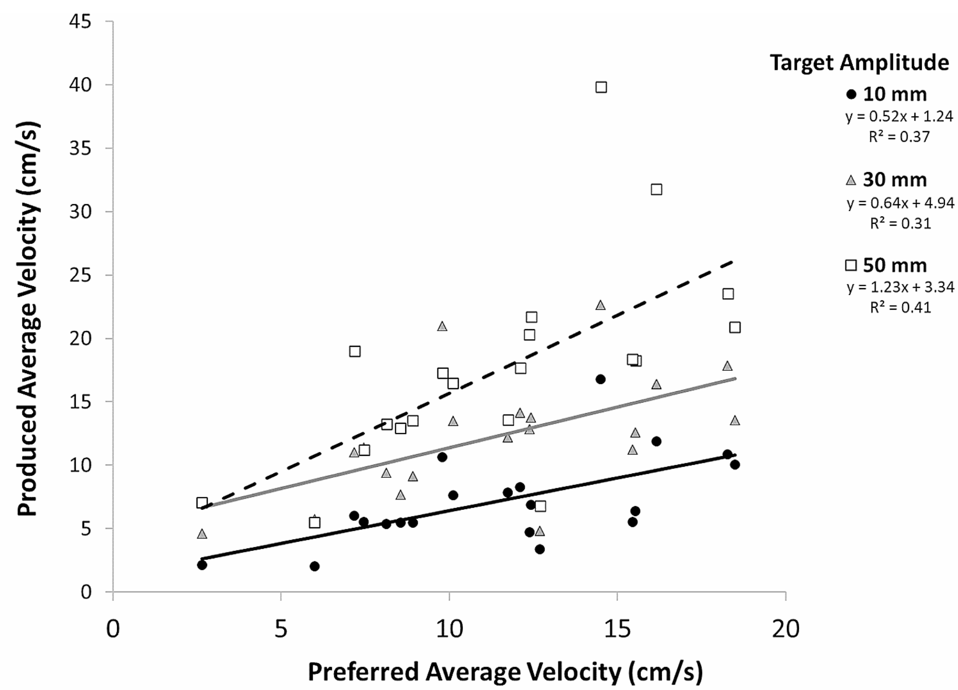


Fig. 6 Scatter plots with best linear fit showing relation between estimated preferred velocities and produced velocities for the three target amplitude conditions. All correlations were significant at Bonferroni corrected $p < 0.017$



velocity (too slow) in the 50 mm target amplitude condition. That is, based on the estimate of preferred velocity from the SMT and SMA tasks, produced inter-tap intervals should have shortened more for the 10 mm target amplitude and lengthened more for the 50 mm target amplitude in order to maintain a single constant preferred velocity.

Discussion

In this study, we investigated the relation between movement amplitude and movement tempo during self-paced (rhythmic) finger tapping using three tasks: a novel spontaneous motor amplitude (SMA) task, the widely studied spontaneous motor tempo (SMT) task, and a target amplitude version of the SMT task that assessed preferred motor tempo at low, medium, and high target finger tapping amplitudes. Of primary interest was whether produced tempi across the three tasks would systematically vary with tapping amplitude with lower and higher tapping amplitudes associated with faster and slower preferred motor tempi, respectively. From this perspective, preferred tempo assessed in the SMT task does not reflect a pure tempo preference, but rather a velocity preference that takes into account the relationship between produced amplitude and produced tempo (expressed as the mean inter-tap interval).

There were three main findings. First, the study is the first that we are aware of to establish preferred movement amplitudes for self-paced rhythmic finger tapping, providing a point of comparison for future studies. We found that for adults, average preferred finger tapping amplitude was ~30 mm, with a range of ~13 to ~48 mm. Notably, preferred amplitude was unrelated to finger length or sex, but was weakly related to maximum possible produced amplitude.

Second, regardless of whether participants performed the novel SMA task, which focused participants' attention on their preferred tapping amplitude, or the SMT task, which focused participants' attention on their preferred tapping tempo, produced amplitudes and tempi were very similar on average and highly correlated across tasks. Moreover, estimated preferred velocities, which combined produced amplitudes and produced inter-tap intervals, were a close match across tasks both when considering task averages and on a participant-by-participant basis. A linear regression examining the relation between estimated velocity in the SMA and SMT tasks accounted for 77% of the variance with a slope not significantly different from 1.

This supports the view that the concepts of preferred tempo and preferred amplitude should not be considered in isolation, but rather are interdependent.

For both the SMA and SMT tasks, the observed range of produced tempi (SMA: 264–1263 ms; SMT: 275–1242 ms) is consistent with the range of preferred tempi reported for adults in previous studies (Collyer et al. 1994; Drake et al. 2000; Fraisse 1982; Hammerschmidt et al. 2021; McAuley et al. 2006; Moelants 2002). The present study suggests that some of this observed individual variation in preferred tempo in self-paced finger tapping studies may reflect individual differences in tapping amplitude associated with a preferred velocity that constrains the relation between the tempo and amplitude measures. If so, then it is possible that faster preferred tempi observed for children compared to adults (Drake et al. 2000; McAuley et al. 2006) may reflect, in part, differences in produced amplitudes, rather than pure differences in preferred tempo.

Finally, for the target amplitude version of the SMT task, participants were found to alter their preferred tempo for the three target amplitudes in the direction predicted by the preferred velocity hypothesis. The lowest, 10 mm, target amplitude, resulted in the fastest preferred tempo, the intermediate, 30 mm, target amplitude resulted in an intermediate preferred tempo, and the highest target amplitude, 50 mm, resulted in the slowest preferred tempo. Although there was individual variation in the direction of preferred tempo shift across target amplitudes, 85% of participants showed a slowing of preferred tempo from the 10 mm to the 50 mm target amplitude condition, altering their preferred tempo with tap amplitude in the direction predicted by their preferred velocity.

A closer examination of the produced preferred tempi at the three target amplitudes shows that predicted preferred tempo based on the estimated velocity from the SMA and SMT tasks matched the produced preferred tempo in the target amplitude version of the SMT task at the intermediate 30 mm target amplitude, but not at the low (10 mm) and high (50 mm) target amplitudes. For the low and high amplitude targets, preferred tempo did not speed up and slow down enough respectively to match the predicted preferred tempo based on the estimated velocity derived from the SMA and SMT tasks. Though velocity was constant across SMA and SMT tasks when participants were tapping at a comfortable amplitude or tempo, velocity changed across target amplitude conditions, indicating that participants did not maintain a constant preferred velocity when required to tap at the extreme high or low end of the amplitude range. Thus, overall, the results from the target amplitude version of the SMT task support the weak, but not the strong, version of the preferred velocity hypothesis.

An additional finding from the target amplitude version of the SMT task was that participants tended to produce amplitudes larger than the target. This may be due to the nature of the target amplitude version of the SMT task, since participants were required to match the target amplitude by visibly disrupting the string on the apparatus, which led them to avoid underestimating target amplitude (i.e., not raising their finger high enough to touch the string). Although several participants reported having some difficulty tapping in the 50 mm target amplitude condition, only one produced a mean tap amplitude below 50 mm in this condition.

The phenomenon of preferred tempo has been studied in a variety of contexts, but arguably most widely investigated using the self-paced finger tapping task examined in the current study. Results of the present study extend previous studies by providing evidence that SMT assessments of preferred tempo do not reflect a time preference independent of spatial movement characteristics. These results highlight the interdependence of movement tempo and movement amplitude in assessments of preferred tempo and are consistent with other work showing a relation between movement amplitude and tempo in paced tasks (Kay et al. 1987; Rosenbaum et al. 1991).

The dependence of tempo preferences on spatial movement characteristics demonstrated in the present study falls in line with previous studies proposing that the SMT task reflects endogenous processes that underly rhythmic movements (McPherson et al. 2018). Tempi in the range of typical adult

SMT have been suggested to reflect an ideal range for both sensorimotor synchronization and neural entrainment by auditory stimuli (Fujioka et al. 2012; Guerin, 2021; Kliger-Amrani and Zion-Golumbic 2020; Miyake et al. 2004; Tierney and Kraus 2013) and neuroimaging studies have demonstrated that rhythms with regular intervals in this range recruit motor areas of the brain even when no movement is executed (Chen et al. 2008; Grahn and Brett 2007). Furthermore, auditory rhythms presented at or near an individual's SMT prime hand movements more readily than faster or slower tempi (Michaelis et al. 2014).

The results of the present study have potential implications for understanding how individuals with movement disorders produce regularly timed movements. Previously, characteristic differences in preferred tempo using the SMT task have been shown in some clinical populations such as individuals with Parkinson's disease (PD). Rose et al. (2020) had adults with PD and healthy controls tap with their finger or toes at their preferred tempo. For finger tapping, patients with PD tended to produce faster SMTs with more variable timing than controls. By demonstrating a link between the spatial and temporal characteristics of tapping movements, the present study raises the possibility that producing faster or more variable inter-tap intervals by individuals with PD in rhythmic tapping tasks may not reflect a timing deficit per se, but rather reflect alterations in the spatial components of these individuals' movements. Individuals with PD have also been shown to produce hypometric movements – movements with smaller and more variable amplitudes compared to age-matched controls (Broderick et al. 2009; Moustafa et al. 2016; Stegemöller et al. 2019; Van Gemmert et al. 1999). Based on the preferred velocity hypothesis, smaller and more variable amplitudes would be expected to lead to faster and more variable produced inter-tap intervals in the SMT task, highlighting the difficulty of separating spatial and temporal effects on movement control. Spatial deficits in turn may give the appearance of a timing deficit.

In sum, this study examined the relation between movement amplitude and tempo in assessments of preferred tempo using the widely used SMT task involving self-paced rhythmic finger tapping. Consistent with the preferred velocity hypothesis, participants (1) produced similar amplitudes and tempi regardless of instructions to produce either their preferred amplitude or preferred tempo maintaining the same average movement velocity across tasks and (2) altered their preferred tempo for different target amplitudes in the direction predicted by their estimated preferred velocity from the SMA and SMT tasks. Overall, our results suggest that the numerous previous assessments of preferred tempo using the SMT task have not provided a pure tempo preference that is independent of movement amplitude. Rather, movement amplitude and preferred tempo are interdependent in the SMT task (and related tasks), suggesting that some inter-individual variation in preferred tempo may be due to differences in the spatial characteristics of individuals' movements.

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Data availability Data are available upon request.

Code Availability Not applicable.

Declarations

Conflicts of interest The authors declare that they have no conflict of interest.

Consent to participate Informed consent was obtained from all individual participants included in the study.

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